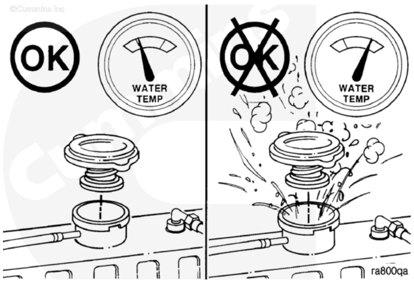
Trainings

Drain

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

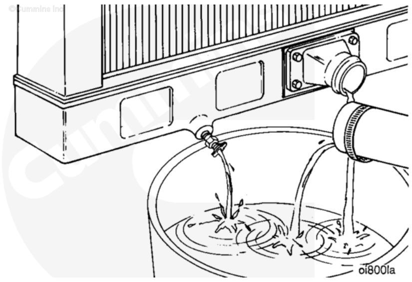


Note : Marine engines will have a heat exchanger or keel coolers instead of a radiator.

Remove the radiator pressure cap after the engine is cool.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



Position the vehicle on level ground.

Open the draincock at the bottom of the radiator.

Remove the lower radiator hose.

Flush

⚠ WARNING ⚠

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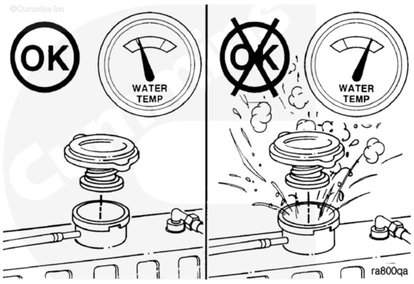


Remove the radiator pressure cap after the engine is cool.

The performance of RESTORE™ cleaning solution is dependent on time, temperature, and concentration levels.

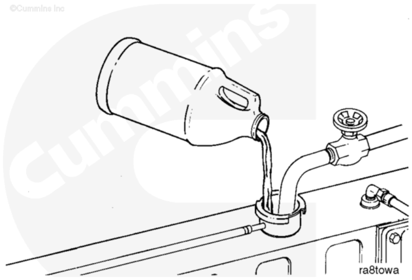
An extremely scaled or flow restricted system can require higher concentrations of cleaners, higher temperatures, or longer cleaning time.

RESTORE™ cleaning solution can be safely used up to twice the recommended concentration levels.

Extremely scaled or fouled systems can require more than one cleaning.

⚠ CAUTION ⚠

Fleetguard RESTORE® cleaning solution does not contain antifreeze. Do not allow the cooling system to freeze during the cleaning operation.



Immediately add 3.8 liters [1 gal] of Fleetguard RESTORE™, cleaning solution or equivalent, for each 38 to 57 liters [10 gal] of cooling system capacity.

Fill the cooling system with plain water.

Turn the heater temperature to HIGH to allow maximum coolant flow through the heater core.

The blower does **not** have to be on.

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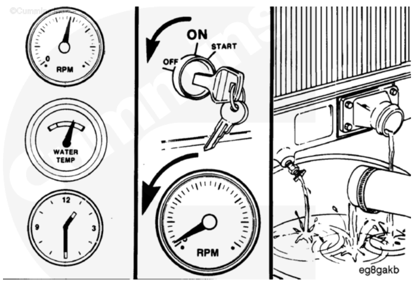
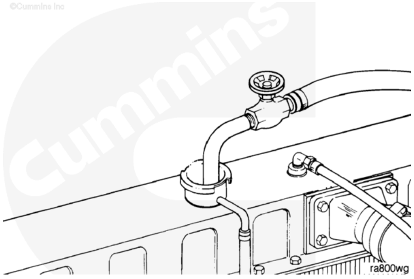
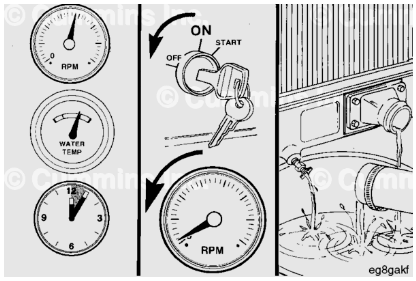


Operate the engine to a minimum coolant temperature of 85°C [185°F] for 1 to 1½ hours.

Shut the engine OFF. Allow it to cool to 50°C [120°F]. Drain the cooling system.

Fill the cooling system with clean water.





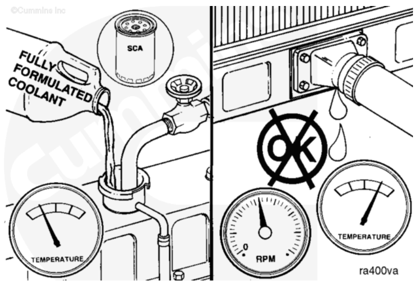
⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

Operate the engine at high idle for five minutes with the coolant temperature above 85°C [185°F].

Shut the engine OFF. Allow it to cool to 50°C [120°F]. Drain the cooling system.

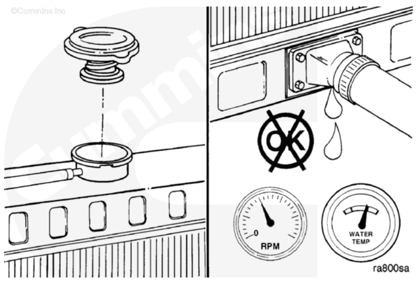
If the water drained is dirty, the system **must** be flushed again until the water is clean.



Install a new coolant filter. Refer to Procedure 008-006 (/qs3/pubsys2/xml/en/procedures/18/18-008-006-tr.html).

Fill the cooling system with fully formulated coolant.

Use additional SCA to bring the coolant to the correct SCA concentration level. Refer to Procedure 008-046 (/qs3/pubsys2/xml/en/procedures/99/99-008-046.html).



Install the pressure cap.

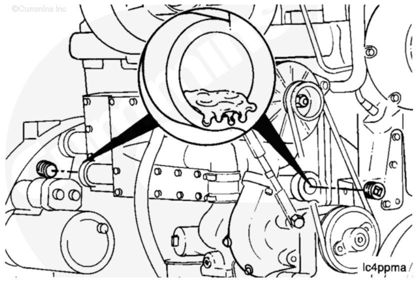
Operate the engine until the coolant reaches a temperature of 70°C [160°F], and check for coolant leaks.

Pressure Test

Internal

Remove the plugs from the oil cooler housing and check for coolant.

If coolant is present, replace the oil cooler elements. Refer to Procedure 007-007 (/qs3/pubsys2/xml/en/procedures/18/18-007-007.html).

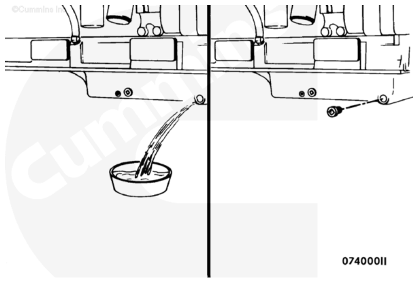


⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused dispose of in accordance with local environmental regulations.



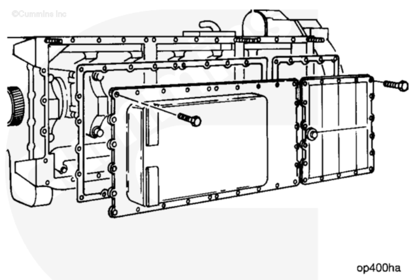
If the oil cooler elements are **not** leaking, drain the oil. Refer to Procedure 007-037 (/qs3/pubsys2/xml/en/procedures/18/18-007-037-tr.html).

Check for coolant in the oil.

If the oil does **not** contain coolant, check the aftercooler. Refer to Procedure 010-008 (/qs3/pubsys2/xml/en/procedures/18/18-010-008.html).

If the oil contains coolant:

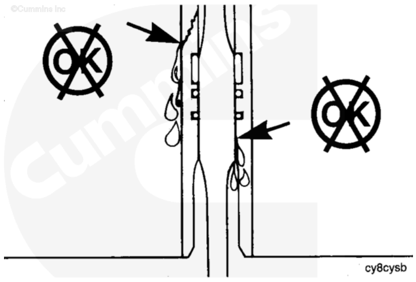
- Remove the lubricating oil pan. Refer to Procedure 007-025 (/qs3/pubsys2/xml/en/procedures/18/18-007-025-tr.html).
- Remove the lubricating oil pan adapter cover. Refer to Procedure 007-026 (/qs3/pubsys2/xml/en/procedures/18/18-007-026-tr.html).



Check for leakage on the inside and outside of the cylinder liners.

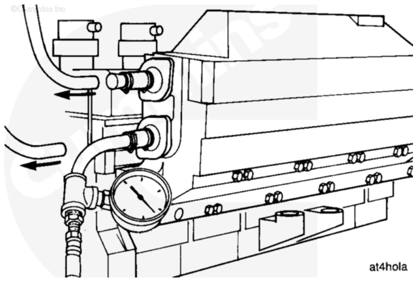
Replace all cylinder liners that are leaking.

If the cylinder liners are **not** leaking, check the aftercooler element.



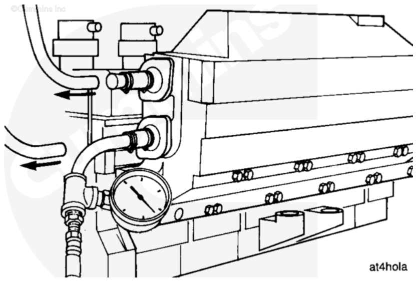
Disconnect the aftercooler coolant hoses. Plug one hose and apply 415 kPa [60 psi] of air pressure.

If the air pressure decreases, replace the aftercooler element. Refer to Procedure 010-008 (/qs3/pubsys2/xml/en/procedures/18/18-010-008.html).



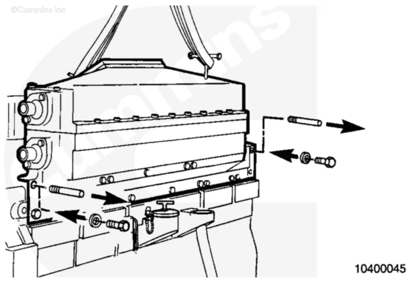
Remove the test equipment. Remove the plug from the overflow tube.

Replace the pressure cap.



If the aftercooler element is **not** leaking, check the cylinder heads.

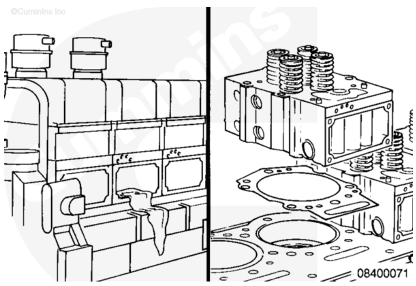
Remove the aftercooler housing. Refer to Procedure 010-002 (/qs3/pubsys2/xml/en/procedures/18/18-010-002-tr.html).



Check the intake ports in the cylinder head for coolant.

If the intake ports have coolant, replace the cylinder head.

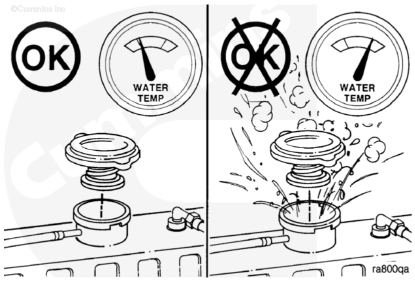
Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/18/18-002-004-tr.html).



External

⚠ WARNING ⚠

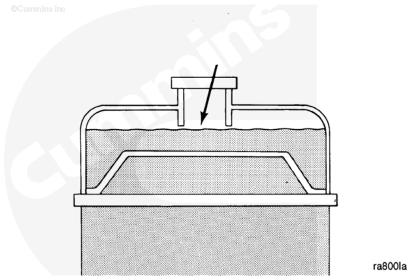
Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



To confirm and locate external coolant leaks, it is **not** necessary to warm the engine.

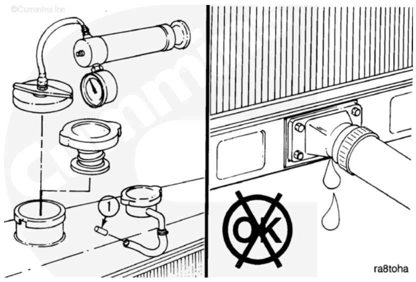
Check the coolant and fill with heavy duty coolant if necessary.

To confirm and locate external coolant leaks that occur during normal operation, complete testing while the engine is warm.



⚠ CAUTION ⚠

Do not apply more than 140 kPa [20 psi] of air pressure to the cooling system. Excessive air pressure can damage the water pump seal.



If the radiator is equipped with a pressure relief valve, install a plug in the overflow tube (1).

Install the pressure tester on the radiator fill neck or surge tank (if equipped).

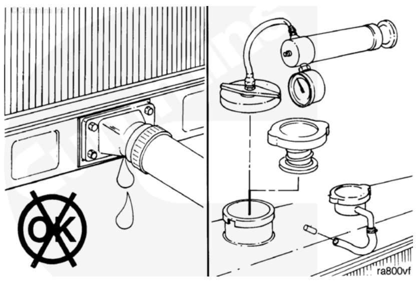
Apply a maximum of 140 kPa [20 psi] air pressure.

Maintain the air pressure for a minimum of five minutes.

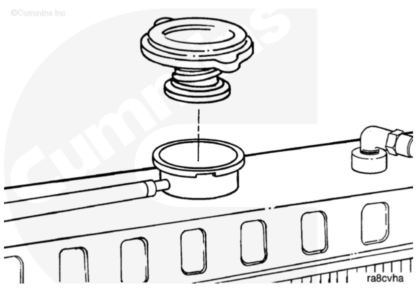
Check for leaks. Any leaks **must** be repaired.

Remove the pressure test equipment.

Remove the plug from the overflow tube (if equipped).

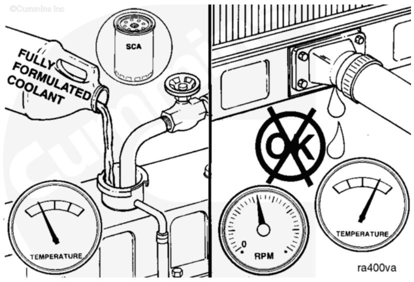


Replace the coolant system pressure cap.



Fill

Close the radiator draincocks.



Install the lower radiator hose(s).

Tighten the hose clamps.

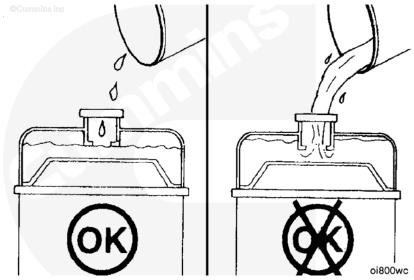
Torque Value: 5 n•m [40 in-lb]

Use fully formulated coolant to fill the coolant system.

Use the correct units of SCA to obtain the correct cooling system protection. Refer to Procedure 008-046 (/qs3/pubsys2/xml/en/procedures/99/99-008-046.html).

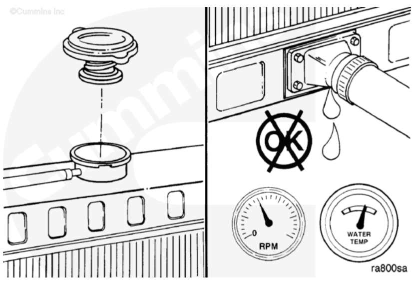
⚠ CAUTION ⚠

If the coolant level is above the bottom of the fill neck there will not be enough space for the air that is in the system. If the air is not trapped at the top of the radiator it can travel to the water pump inlet causing low coolant flow because of impeller cavitation. The cavitation can result in overheating of the engine.



Fill the cooling system with coolant to the bottom of the fill neck in the radiator or expansion tank.

Replace the radiator or fill cap.



Operate the engine until the coolant reaches a temperature of 70°C [160°F].

Check for leaks.

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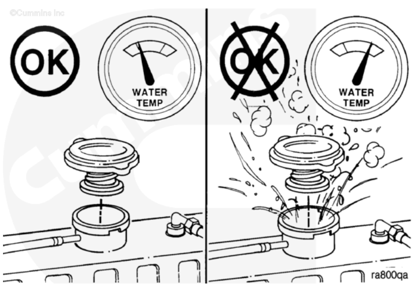

Shut the engine off, allow it to cool and check the coolant level.